INSPEC TESTING SERVICES

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Test Report

ANSI Z359.12-2019 Connecting Components for Personal Fall Arrest Systems

Report no:	2.24.03.03
Customer:	Ninghai Jinsong Stamping Co., Ltd. Shangpu Industrial Park Qiangjiao Town, Ninghai Ningbo City, Zhejiang China
Manufacturer: **as advised by the Customer**	Ninghai Jinsong Stamping Co., Ltd.
Customer order:	T/1270B
Order received:	29 Nov. 2023
Model:	JS-3009
Dates of tests:	11 Dec. 2023 to 2 Mar. 2024

Signed:

Steven Sum, Laboratory Manager

Issued: 5 March 2024

Conditions

This report may be reproduced and distributed to your customers, provided that it is reproduced and distributed in full.

Specimens will be disposed of four weeks from the date of this report, unless otherwise instructed.

Opinions, comments and interpretations expressed in this report are shown in italics.

Copies of INSPEC interpretations referenced in this report are available upon request.

Tests marked ☒ are not included in our ANAB Scope of Accreditation.

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Summary of assessment *

Clause	Requirement	Assessment (See Key)
3.1.1	Surface finish of hardware	**Pass**
3.1.2	New and Unused	**Pass**
3.1.3	Snaphooks & Carabiners	
3.1.4	D-rings, O-rings & Oval rings	**Pass**
3.1.5	Buckles and adjusters	
3.1.6	Proof load testing	
3.1.7	Dynamic drop test	**Pass**
5.1 / 5.2	Marking	**Ltd**
5.3	Instructions	

Key

	Shading shows the clauses requested. Any other clauses were not requested.
Pass	Requirement satisfied.
Ltd	Testing requested was insufficient completely to verify compliance with the clause. Refer to the "Result details" section for more information.
Fail	Requirement not satisfied. Refer to the "Result details" section for more information.
NAs	Assessment not carried out.
NAp	Requirement not applicable.
NT	Requested but not tested due to early termination following failure.

* Assessment relates only to those specimens which were tested and are the subject of this report.

Submission details

Product	Quantity	Date received	INSPEC specimen no. (2L383+)
D-ring (semi-circle), model JS-3009	09	7 December 2023	01 to 09

Procedures

The specimens detailed within the submission above were used for the tests covered by this report.

Testing was performed in accordance with ANSI Z359.12-2019 unless otherwise specified below. Reference should be made to the standard when reading this report.

Unless stated otherwise, specimens were tested in the condition as received by INSPEC.

Testing was performed at INSPEC's laboratory in Kunshan City, China.

Result details

3 Requirements

3.1 Component and Element Requirements

3.1.1 Surface Finish of Hardware

Specimens 2L38301, 2L38302 and 2L38303 were assessed.

a) The finish of the specimens were clean and free of scale, rust and deposits of foreign matter. **Pass**

b) Following the salt spray test, there was no evidence of either, red rust visible to the unaided eye, or corrosion of the base metal of the specimens. **Pass**

c) All surfaces of the specimens, which may come in contact with tearable materials, were free of burrs, pits, sharp edges and rough surfaces. **Pass**

3.1.2 Condition of Hardware

All specimens were new and unused when received. **Pass**

3.1.4 D-rings, O-rings and Oval rings

Specimens 2L38304, 2L38305 and 2L38306 withstood the 5,000 lb (22.2kN) tensile load applied for 1-minute without breaking. **Pass**

3.1.7 Dynamic drop test

Following abrasion and cold conditioning, specimens 2L38307, 2L38308 and 2L38309 withstood the dynamic drop tests without permanent deformation. **Pass**

5.1 General Marking Requirements

Markings were provided electronically and use for assessment.

5.1.1 Markings shall be in English.

5.1.2 The requirement that markings shall be expected to remain present and legible throughout the expected life of the component being marked was not assessed. **NAs**

5.1.3 Any restrictions on the use of such connectors (hardware) shall be marked on the connectors (hardware) or components, subsystems and systems of which they are an integral part. [Nil] **NAp**

5.2 Specific Marking Requirements

5.2.1 Connectors. Connectors (all components listed in 3.1.6, requiring proof loading as per this standard, i.e. Snaphooks, Carabiners, D-rings, O-rings and Oval rings) shall be marked to identify the following:

· Year of manufacture; [2023] **Pass**

· Manufacturer's identification; [logo] **Pass**

· Markings for connectors shall be sufficient to provide traceability; [2312] **Pass**

· Load rating for the major axis of the connector stamped or otherwise permanently marked on the device, minimum 5,000 lb. (22kN); [5000 lb] **Pass**

· Load rating for gate stamped or otherwise permanently marked on the gate mechanism; **NAp**

· For connectors that are non-integral part (non-captive eye), then "ANSI Z359.12-XX" ("XX" denotes year of standard) is required. **NAp**

Estimates of the uncertainty of measurement

Clause	Test		Uncertainty
3.1.1	Surface finish of hardware		See Note 1
3.1.2	New and Unused		Not applicable
3.1.3	Snaphooks and carabiners	Tensile test	See Note 1
		Gate resistance	See Note 1
		Gap measurement	±1.4%
3.1.4	D-rings, O-rings and Oval rings		See Note 1
3.1.5	Buckles and Adjusters		See Note 1
3.1.6	Proof load testing		See Note 1
3.1.7	Dynamic Drop test		See Note 1
5	Marking and Instructions		Not applicable

Note 1 The acceptance criterion for this test is a straightforward “Pass/Fail”, rather than a numerical value. Consequently, as there is no value to be reported, uncertainty has not been reported either.

Note 2 The uncertainty value is based on a standard uncertainty multiplied by a coverage factor k = 2, which provides for a confidence level of approximately 95%. Values expressed as a percentage (%) are relative.

Note 3 It should be noted that the above values have not been taken into account when making assessment to the pass/fail criteria.

ANNEX

This Annex comprises one section.

1. Photograph of the product tested. (1 page)

END OF REPORT

Ninghai Jinsong Stamping Co., Ltd. –
D-ring (semi-circle), model JS-3009

